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Laporan Akhir Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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1 Langkah-Langkah Percobaan

1.1 Praktikum 1: Konfigurasi Cisco Switch (VLAN dan Trunking)

Pada praktikum ini dilakukan konfigurasi VLAN pada switch Cisco serta pengaturan trunk untuk menghubungkan antar switch.

1. Verifikasi konfigurasi VLAN pada switch.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	Finance	active	
20	HR	active	
30	IT	active	
40	Marketing	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Gambar 1: Show VLAN Brief pada Switch

2. Verifikasi konfigurasi trunk antar switch.

```
Switch#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gi0/0	10,20,30,40

Port	Vlans allowed and active in management domain
Gi0/0	10,20,30,40

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	none

Gambar 2: Show Interfaces Trunk

3. Menyimpan konfigurasi perangkat.

```
Switch#write memory
Building configuration...
Compressed configuration from 3364 bytes to 1652 bytes[OK]
Switch#
*Jun  5 08:38:52.282: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun  5 08:38:52.982: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
```

Gambar 3: Save Configuration (Write Memory)

1.2 Praktikum 2: Konfigurasi Cisco Router (ROAS dan DHCP)

Pada praktikum ini dilakukan konfigurasi router untuk inter-VLAN routing dan DHCP server.

1. Verifikasi konfigurasi interface router.

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset    administratively down down
GigabitEthernet0/1    unassigned      YES unset    administratively down down
GigabitEthernet0/2    unassigned      YES unset    up          up
GigabitEthernet0/2.10 192.168.10.1    YES manual   up          up
GigabitEthernet0/2.20 192.168.20.1    YES manual   up          up
GigabitEthernet0/2.30 192.168.30.1    YES manual   up          up
GigabitEthernet0/2.40 192.168.40.1    YES manual   up          up
GigabitEthernet0/3    unassigned      YES unset    administratively down down
GigabitEthernet0/4    unassigned      YES unset    administratively down down
```

Gambar 4: IP Interface Brief Router

2. Konfigurasi DHCP pool.

```
Pool VLAN10-FINANCE :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.10.12 192.168.10.1 - 192.168.10.254 1

Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1
```

Gambar 5: DHCP Pool Configuration

3. Verifikasi DHCP binding.

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
Hardware address/
User name
192.168.10.11    0100.5079.6668.6e Jun 06 2026 08:51 AM Automatic
192.168.20.11    0100.5079.6668.6d Jun 06 2026 08:51 AM Automatic
```

Gambar 6: DHCP Binding Table

4. Verifikasi IP client.

```

VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 192.168.10.11/24
GATEWAY        : 192.168.10.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.10.1
DHCP LEASE     : 86392, 86400/43200/75600
MAC            : 00:50:79:66:68:6e
LPORT          : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500

```

Gambar 7: IP Address Client

5. Pengujian gateway VLAN.

<pre> VPCS> ping 192.168.10.1 84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.208 ms 84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.336 ms 84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.133 ms 84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.763 ms 84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.018 ms </pre>	<pre> VPCS> ping 192.168.20.1 84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.000 ms 84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.293 ms 84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.633 ms 84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.136 ms 84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=2.776 ms </pre>
<pre> VPCS> ping 192.168.30.1 84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.925 ms 84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.565 ms 84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=3.035 ms 84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=5.439 ms 84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.268 ms </pre>	<pre> VPCS> ping 192.168.40.1 84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.703 ms 84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.404 ms 84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=3.323 ms 84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.380 ms 84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=3.613 ms </pre>

Gambar 8: Ping Gateway VLAN

6. Pengujian komunikasi antar VLAN.

```

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=5.048 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=4.911 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=5.151 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=5.608 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=4.899 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=4.810 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.967 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=4.851 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.641 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=5.020 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=4.762 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=3.950 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.696 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=5.910 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=4.102 ms

```

Gambar 9: Inter-VLAN Communication Test

7. Verifikasi DHCP binding akhir.

```

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.6e    Jun 06 2026 08:53 AM    Automatic
192.168.20.11    0100.5079.6668.6d    Jun 06 2026 08:54 AM    Automatic

```

Gambar 10: Final DHCP Binding

1.3 Praktikum 3: Multi-Vendor Networking

Pada praktikum ini dilakukan konfigurasi jaringan antar perangkat Cisco, Huawei, dan MikroTik.

1. Verifikasi IP masing-masing perangkat.

```

VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 192.168.50.10/24
GATEWAY        : 192.168.50.1
DNS            :
MAC            : 00:50:79:66:68:76
LPORT          : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500

```

Gambar 11: IP Configuration Overview

2. Pengujian ping antar jaringan.

```
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=9.810 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.739 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=4.161 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.166 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.408 ms
```

Gambar 12: Ping Gateway Multi-Vendor

3. Verifikasi interface Cisco.

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0       10.10.11.1      YES manual up            up
GigabitEthernet0/1       10.10.30.1      YES manual up            up
GigabitEthernet0/2       unassigned      YES unset  up            up
GigabitEthernet0/2.10    192.168.10.1    YES manual up            up
GigabitEthernet0/2.20    192.168.20.1    YES manual up            up
GigabitEthernet0/2.30    192.168.30.1    YES manual up            up
GigabitEthernet0/2.40    192.168.40.1    YES manual up            up
GigabitEthernet0/3       10.10.10.1      YES manual up            up
GigabitEthernet0/4       unassigned      YES unset  administratively down down
```

Gambar 13: Cisco Interface Brief

4. Verifikasi interface Huawei.

```
[~HUAWEI]commit
[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                IP Address/Mask  Physical  Protocol  VPN
Ethernet1/0/0            10.10.10.2/30    up        up        --
Ethernet1/0/0.4094       128.1.138.137/16 up        up        l3vpn
Ethernet1/0/1            unassigned       up        down      --
Ethernet1/0/1.4094       128.1.138.137/16 up        up        l3vpn
Ethernet1/0/2            192.168.50.1/24  up        up        --
Ethernet1/0/2.4094       128.1.138.137/16 up        up        l3vpn
Ethernet1/0/3            10.10.20.1/30    up        up        --
Ethernet1/0/3.4094       128.1.138.137/16 up        up        l3vpn
Ethernet1/0/4            10.10.11.2/30    up        up        --
Ethernet1/0/4.4094       128.1.138.137/16 up        up        l3vpn
---- More ----
```

Gambar 14: Huawei Interface Brief

5. Konfigurasi IP detail.

```
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
#   ADDRESS          NETWORK      INTERFACE
0   ;;; TO-HUAWEI
    10.10.20.2/30     10.10.20.0  ether1
1   ;;; TO-CISCO
    10.10.30.2/30     10.10.30.0  ether2
[admin@MikroTik] >
```

Gambar 15: IP Address Configuration Print

6. Pengujian konektivitas antar vendor.

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/4 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/3 ms
Router#
```

```
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms
```

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME STATUS
0 10.10.20.1                            56 255 3ms
1 10.10.20.1                            56 255 1ms
2 10.10.20.1                            56 255 1ms
3 10.10.20.1                            56 255 1ms
4 10.10.20.1                            56 255 1ms
5 10.10.20.1                            56 255 2ms
6 10.10.20.1                            56 255 1ms
7 10.10.20.1                            56 255 1ms
sent=8 received=8 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=3ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME STATUS
0 10.10.30.1                            56 255 2ms
1 10.10.30.1                            56 255 1ms
2 10.10.30.1                            56 255 4ms
3 10.10.30.1                            56 255 6ms
4 10.10.30.1                            56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=2ms max-rtt=6ms
```

Gambar 16: Multi-Vendor Connectivity Test

1.4 Praktikum 4: OSPF Multi-Vendor Routing

Pada praktikum ini dilakukan konfigurasi OSPF antar perangkat Cisco, Huawei, dan MikroTik.

1. Verifikasi routing protocol.

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

```

Gambar 17: IP Routing Protocols

2. Verifikasi OSPF neighbor.

```

Router#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
8.3.3.3	1	FULL/BDR	00:00:37	10.10.30.2	GigabitEthernet0/1
2.2.2.2	1	FULL/BDR	00:00:35	10.10.11.2	GigabitEthernet0/0
2.2.2.2	1	FULL/BDR	00:00:32	10.10.10.2	GigabitEthernet0/3

Gambar 18: OSPF Neighbor Cisco

3. Routing table OSPF.

```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

```

public_Routing Table : OSPF						
Destinations : 9		Routes : 14				
OSPF routing table status : <Active>						
Destinations : 5		Routes : 10				
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.10.30.0/30	OSPF	10	110	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	110	D	10.10.10.1	Ethernet1/0/0
192.168.10.0/24	OSPF	10	11	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	11	D	10.10.10.1	Ethernet1/0/0
192.168.20.0/24	OSPF	10	11	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	11	D	10.10.10.1	Ethernet1/0/0
192.168.30.0/24	OSPF	10	11	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	11	D	10.10.10.1	Ethernet1/0/0
192.168.40.0/24	OSPF	10	11	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	11	D	10.10.10.1	Ethernet1/0/0
OSPF routing table status : <Inactive>						
Destinations : 4		Routes : 4				
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.10.10.0/30	OSPF	10	10		10.10.10.2	Ethernet1/0/0
10.10.11.0/30	OSPF	10	10		10.10.11.2	Ethernet1/0/4
10.10.20.0/30	OSPF	10	100		10.10.20.1	Ethernet1/0/3
192.168.50.0/24	OSPF	10	1		192.168.50.1	Ethernet1/0/2

Gambar 19: OSPF Routing Table

4. Pengujian routing antar jaringan.

<pre>VPCS> ping 192.168.50.10 84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=11.543 ms 84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.393 ms 84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.935 ms 84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.005 ms 84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.528 ms</pre>	<pre>VPCS> ping 192.168.50.10 84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.700 ms 84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.413 ms 84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.865 ms 84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.170 ms 84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.501 ms</pre>
<pre>VPCS> ping 192.168.50.10 84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.571 ms 84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.382 ms 84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.613 ms 84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.063 ms 84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.222 ms</pre>	<pre>VPCS> ping 192.168.50.10 84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.021 ms 84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.448 ms 84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.928 ms 84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.385 ms 84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.447 ms</pre>

Gambar 20: Inter-VLAN to Server Ping Test via OSPF

5. Pengujian full connectivity.

```
VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=10.664 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=6.840 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=5.232 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=6.193 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=5.597 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=14.587 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=6.830 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=5.803 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=6.150 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=8.657 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=11.039 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=7.986 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=6.669 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=7.409 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=5.744 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=5.595 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=6.390 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=7.924 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=5.676 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=5.837 ms
```

Gambar 21: Full Network Connectivity Test

1.5 Praktikum 5: OSPF Failover Testing

Pada praktikum ini dilakukan pengujian failover pada jaringan OSPF.

1. Pengujian jalur aktif jaringan.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.732 ms  5.263 ms  3.289 ms
 2  10.10.10.2     8.361 ms  4.759 ms  4.434 ms
 3  *192.168.50.10 11.030 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 22: Active Path Traceroute

2. Pengujian saat salah satu link mati.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.326 ms 5.060 ms 3.293 ms
 2 10.10.11.2 6.736 ms 4.479 ms 4.193 ms
 3 *192.168.50.10 5.529 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 23: Failover Test 1

3. Pengujian failover lanjutan.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.090 ms 2.903 ms 3.369 ms
 2 10.10.30.2 6.229 ms 5.606 ms 2.770 ms
 3 10.10.20.1 8.770 ms 5.739 ms 6.296 ms
 4 *192.168.50.10 11.768 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 24: Failover Test 2

4. Pengujian seluruh link aktif.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.377 ms 4.617 ms 5.323 ms
 2 10.10.10.2 5.596 ms 4.418 ms 5.219 ms
 3 *192.168.50.10 9.551 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 25: Full Connectivity Restored

2 Analisis Hasil Percobaan

Hasil percobaan menunjukkan bahwa seluruh konfigurasi jaringan berhasil dilakukan sesuai dengan tujuan praktikum, mulai dari VLAN, routing, hingga pengujian failover jaringan.

2.1 Praktikum 1: VLAN dan Trunking

Pada praktikum ini, VLAN berhasil dikonfigurasi pada switch Cisco dengan pembagian VLAN 10 dan VLAN 20. Hasil pengujian menunjukkan bahwa perangkat dalam VLAN yang sama dapat saling berkomunikasi dengan baik, sedangkan komunikasi antar VLAN tidak dapat dilakukan. Hal ini sesuai dengan konsep dasar VLAN yang memisahkan broadcast domain secara logis.

Selain itu, konfigurasi trunk antar switch berhasil mengizinkan VLAN 10 dan VLAN 20 untuk dilewatkan melalui satu link fisik menggunakan encapsulation IEEE 802.1Q.

2.2 Praktikum 2: DHCP dan Inter-VLAN Routing

Pada praktikum ini, router berhasil dikonfigurasi untuk memberikan alamat IP secara otomatis menggunakan DHCP. Hasil pengujian menunjukkan bahwa client mendapatkan IP sesuai dengan pool yang telah ditentukan.

Pengujian konektivitas juga menunjukkan bahwa perangkat dalam jaringan yang berbeda dapat saling berkomunikasi setelah routing diaktifkan, yang membuktikan bahwa inter-VLAN routing berjalan dengan baik.

2.3 Praktikum 3: Multi-Vendor Networking

Hasil percobaan menunjukkan bahwa perangkat dari vendor yang berbeda (Cisco, Huawei, dan MikroTik) dapat saling terhubung. Hal ini membuktikan bahwa protokol TCP/IP memungkinkan interoperabilitas antar perangkat yang berbeda selama konfigurasi IP address dilakukan dengan benar.

2.4 Praktikum 4: OSPF Routing

Pada praktikum ini, protokol OSPF berhasil membentuk adjacency antar router dan membangun Link State Database (LSDB). Hasil pengujian routing menunjukkan bahwa setiap router dapat menentukan jalur terbaik menggunakan algoritma Shortest Path First (SPF).

Routing table juga menunjukkan bahwa rute OSPF berhasil terinstal secara otomatis, sehingga komunikasi antar jaringan dapat dilakukan dengan efisien.

2.5 Praktikum 5: Failover OSPF

Pengujian failover menunjukkan bahwa ketika salah satu link jaringan dimatikan, OSPF secara otomatis menghitung ulang jalur alternatif. Hal ini dibuktikan dengan hasil traceroute yang berubah sesuai dengan kondisi jaringan.

Ketika semua link aktif kembali, jaringan kembali memilih jalur terbaik secara otomatis tanpa perlu konfigurasi tambahan.

3 Hasil Tugas Modul

Tugas modul terlampir di README.md

4 Kesimpulan

Berdasarkan hasil praktikum yang telah dilakukan, dapat disimpulkan bahwa konfigurasi jaringan menggunakan VLAN, trunking, routing, DHCP, dan OSPF berhasil diimplementasikan sesuai dengan tujuan praktikum.

Pada praktikum VLAN, perangkat dalam VLAN yang sama dapat saling berkomunikasi, sedangkan komunikasi antar VLAN tidak dapat dilakukan tanpa adanya routing. Hal ini membuktikan bahwa VLAN berhasil memisahkan broadcast domain secara logis.

Pada praktikum routing dan DHCP, perangkat berhasil memperoleh alamat IP secara otomatis serta dapat berkomunikasi antar jaringan melalui router. Selain itu, interkoneksi antar perangkat dari vendor berbeda juga berhasil dilakukan dengan baik.

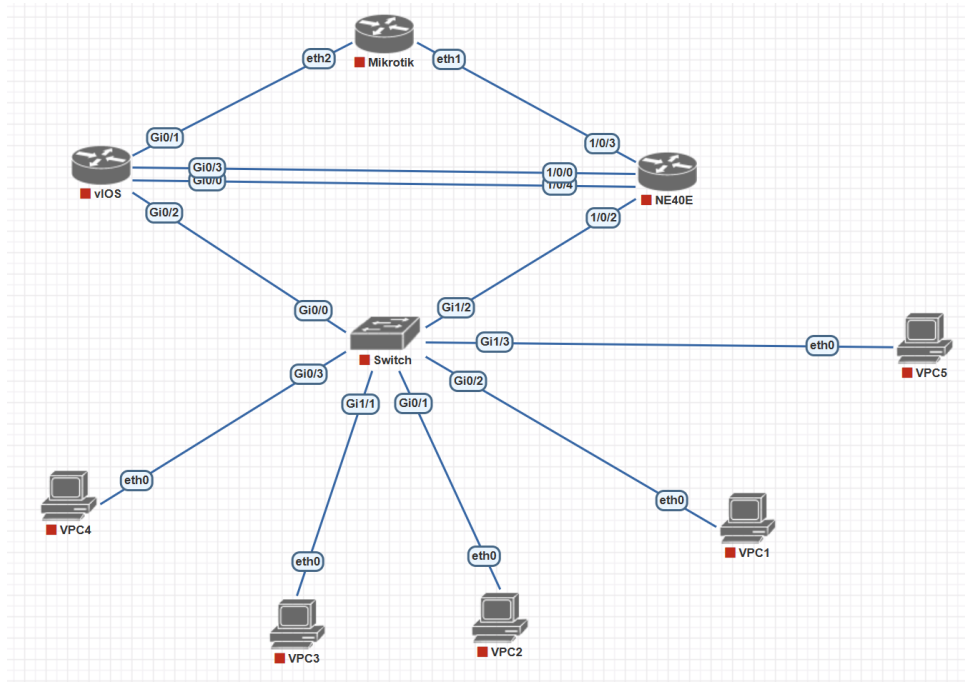
Pada praktikum OSPF, protokol routing dinamis berhasil membentuk tetangga (neighbor), membangun tabel routing, dan menentukan jalur terbaik menggunakan algoritma Shortest Path First (SPF). Hal ini menunjukkan bahwa OSPF mampu mengelola routing secara otomatis dan efisien.

Terakhir, pada pengujian failover, OSPF mampu melakukan penyesuaian jalur secara otomatis ketika terjadi perubahan pada topologi jaringan, sehingga konektivitas tetap terjaga.

Secara keseluruhan, seluruh hasil praktikum telah sesuai dengan teori yang dipelajari dan memberikan pemahaman yang lebih mendalam mengenai konsep jaringan komputer modern.

5 Lampiran

5.1 Dokumentasi saat praktikum



Gambar 26: Topologi Jaringan Praktikum